

FIT5145 - A3 Presentation

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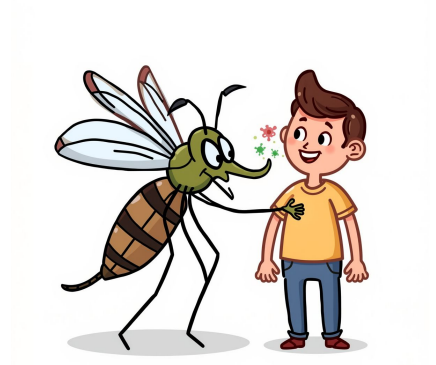
1. Current issues

The Crisis: Escalating climate changes shift mosquito breeding behaviors, causing unpredictable multi-disease outbreaks

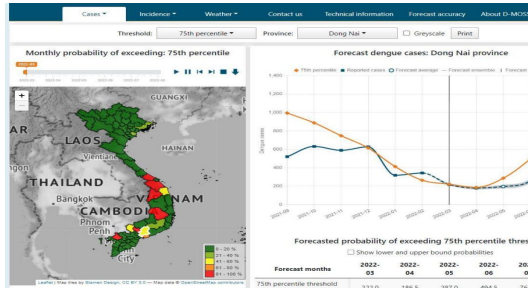
The Gap: Existing systems look at environmental factors or clinical records in isolation, completely missing localized operational healthcare deficits

Project Objective: Build an integrated forecasting model to predict multiple dangerous mosquito-borne disease and stop the spread.

Image source: Canva . (2026). CanvaAI (May 24 version) [LLM]
<https://www.canva.com/ai>

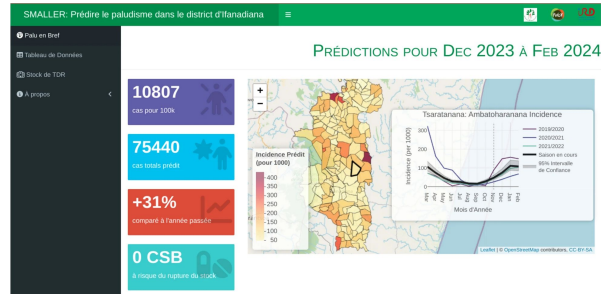


2. Related work



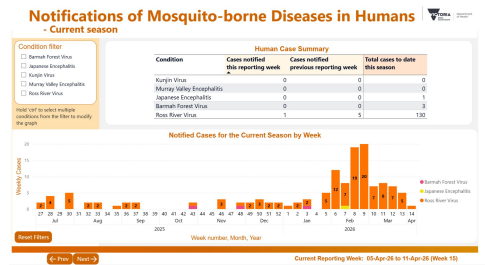
D-MOSS: Use in Vietnam, Malaysia, Sri Lanka. Forecast **dengue** fever 6 months early by collecting environmental and local historical data.

Gap: Focusing only a single disease.



MEWS: Use in Africa and India to predict the probability of **malaria** happening by collecting environmental data.

Gap: Focusing only a single disease and only consider environmental factors



Victoria surveillance report:

Use in Victoria state, determine existed disease from trapped mosquito.

Gap: Not real-time information, not consider environmental factors.



3. Business model

Business area: Sits within the data science, epidemiology, and public health informatics domains.

Data handling: Collect from external source and store them on cloud. Use ETL process to integrate data.

Benefits and values: Medical, financial, environmental benefits.

Stakeholders: local residents, government, healthcare professionals, and environmentalists.

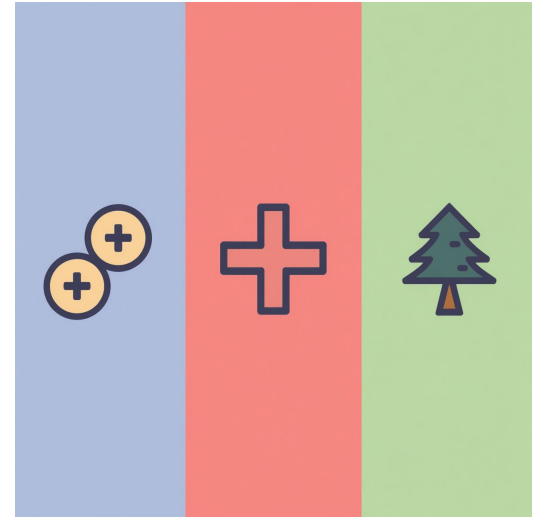


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4.1. NIST Framework

Data Source: Weather stations, satellites, hospitals, and labs.

Volume: Huge datasets (gigabytes per year) covering provinces of 195+ countries.

Velocity: Mixes fast, daily weather updates with slower, monthly hospital records.

Variety: Structured, semi-structured, and unstructured data.

Veracity: Cleans up data errors like broken sensors, bad mosquito counts, and human typos.

Software: AWS, Apache tools (Storm/Hive/Spark), Watson, Python, Rstudio.

Analytics: Shows past trends (descriptive) and forecasts future outbreaks (predictive).

Processing: Takes in live daily streams, processes large monthly batches, and retrains machine learning models.

Capabilities: Merges different data types to find patterns and send real-time risk alerts.

Security/Privacy: Strictly protects patient privacy and keeps medical records anonymous.

Lifecycle: Safely archives old historical data and adapts the system when incoming data formats change.

Other: Flexible enough to handle different medical reporting rules across different countries.



4.2. Data analysis and statistical methods

Multiple Linear Regression (MLR): A baseline model used to measure the exact, direct impact of individual factors.

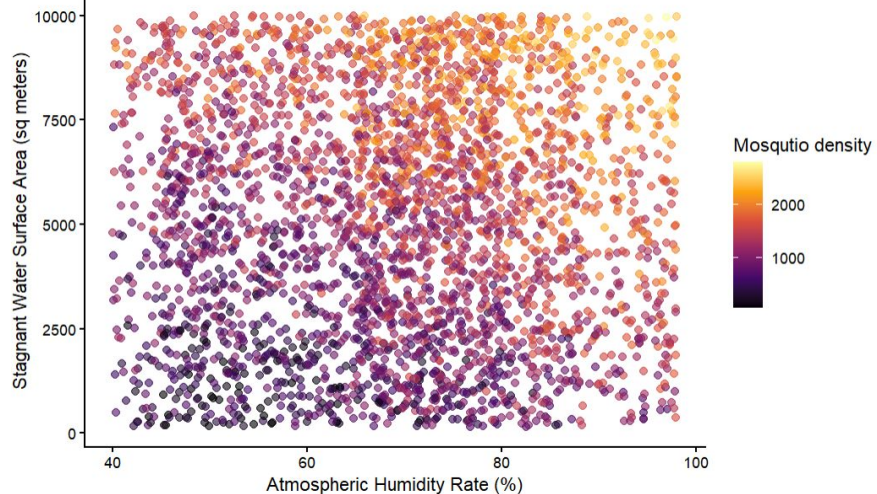
Long Short-Term Memory (LSTM): A neural network that learns from historical time-series data to capture delayed effects.

Multilayer Perceptron (MLP): A deep learning model that uncovers complex, hidden patterns by combining different social and environmental datasets.

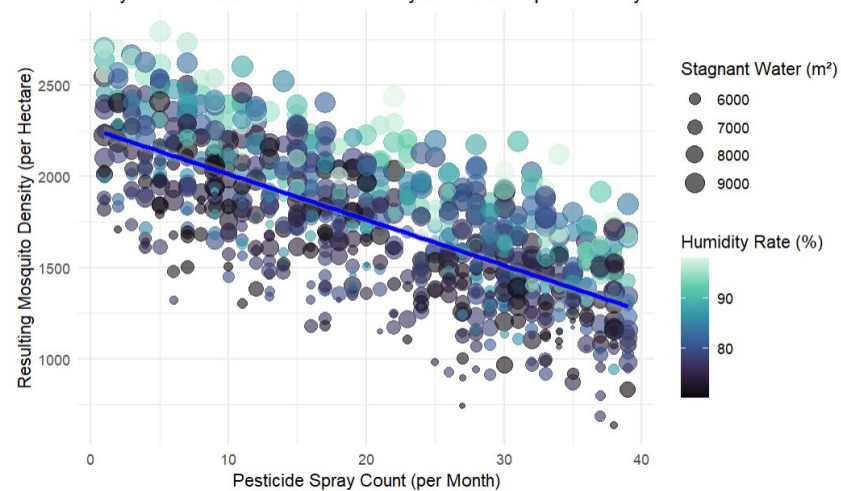
K-Means Clustering: An unsupervised algorithm that automatically groups different health regions into clear risk levels based on shared vulnerabilities.

4.3. Demonstration

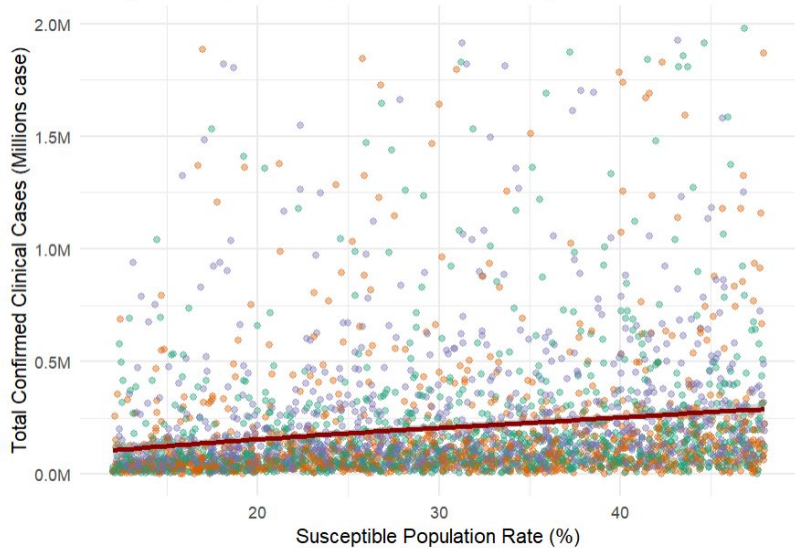
Analysis 1: Mosquito density Distribution across Humidity & Stagnant Water



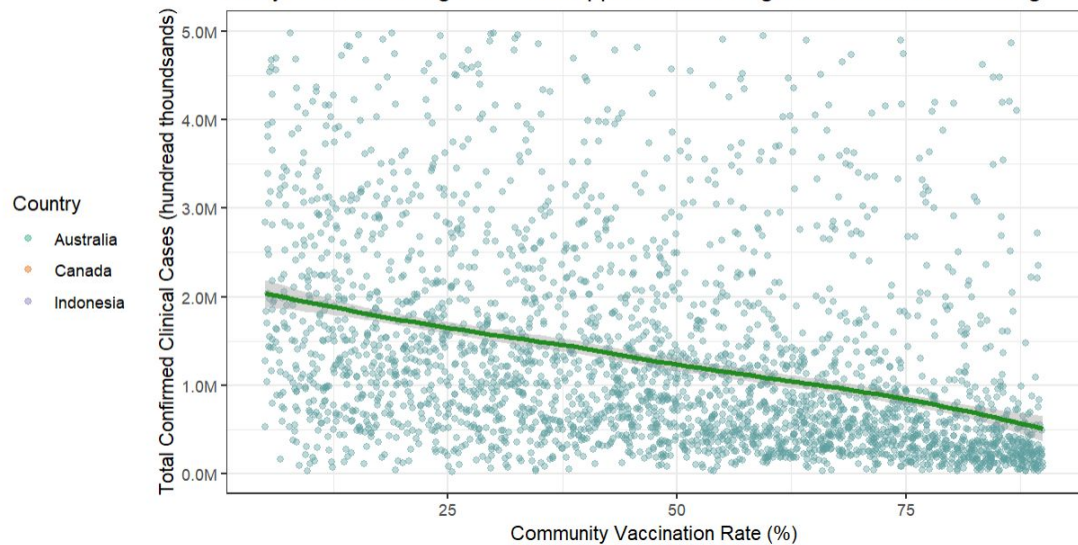
Analysis 2: Pesticide Control Efficacy Within Mosquito density rate



Analysis 3: Impact of Population Susceptibility on Case Counts



Analysis 4: Evaluating Outbreak Suppression via Regional Vaccination Coverage



A tibble: 9 × 5

Country <chr>	Province <chr>	Average_cases <dbl>	Average_Bed_Capacity <dbl>	Bed_Deficit_Ratio <dbl>
Indonesia	Jakarta	440360.3	6126.856	71.86206
Indonesia	West Java	390210.2	5914.786	65.96084
Canada	Quebec	391126.6	6147.669	63.61158
Canada	British Columbia	334738.7	5752.031	58.18477
Australia	Victoria	354029.6	6187.797	57.20492
Indonesia	Bali	341260.1	6175.678	55.24978
Australia	Queensland	329738.2	6269.567	52.58507
Australia	New South Wales	322080.3	6153.350	52.33376
Canada	Ontario	292944.8	6088.022	48.11032

9 rows



5.1. Data Science Process

Plan: Define data requirements and secure sharing agreements.

Collect: Ingest weather, hospital, and mosquito lab data.

Assure: Filter sensor and human errors with strict quality control.

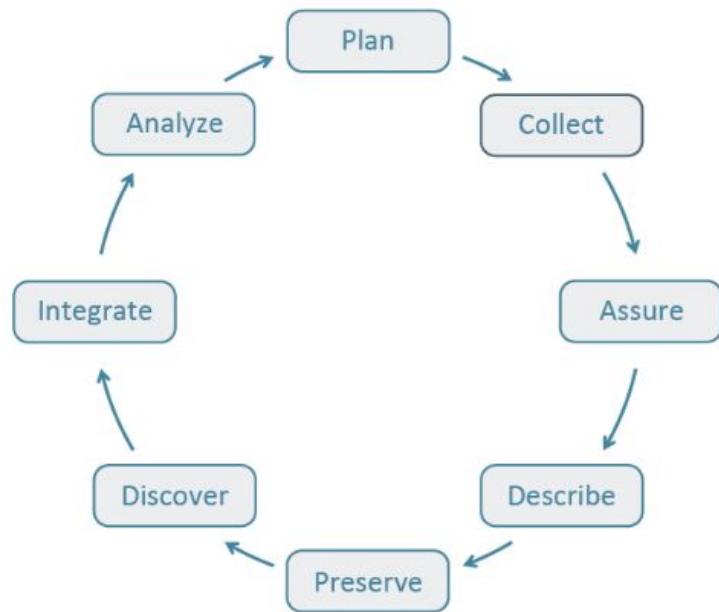
Describe: Document data logic, metrics, and definitions.

Preserve: Securely archive historical data in the cloud (AWS).

Discover: Enable quick data retrieval using metadata tags.

Integrate: Merge diverse datasets using automated ETL pipelines.

Analyze: Forecast outbreaks using Machine Learning (Spark/Python).





5.2. Data Governance and Management

Data Accessibility: Public dashboards are open to everyone, but raw data is restricted.

eDiscovery: Uses metadata tags for fast data retrieval.

Security: Using end-to-end encryption and Role-Based Access Control (RBAC)

Automated Validation: Automatically detects and removes missing values and incorrect data types.

Data Archiving: Compresses and securely stores records older than 20 years.

Privacy & Ethics: Protects patient identities by completely removing Personally Identifiable Information (like names and IDs).

Smooth Operations: Sets clear rules for team roles, system maintenance, and crisis management.